



SIH 26027

AI-Powered Automatic Block Planning for Indian Railways

Smart India Hackathon 2026

Ministry of Railways

Complete Project Guide

Competitor Analysis

Winning Strategy

Prepared: September 7, 2026 | For Team Reference Only

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1. What Does the Problem Statement Actually Ask?

⚠️ IMPORTANT

Read this section carefully — every feature we build must trace back to these 4 requirements.

The PS asks for **4 specific things**:

#	PS REQUIREMENT	WHAT IT ACTUALLY MEANS
1	Integrate maintenance data (defects, overdue) from TMS, SMMS, TDMS with corridor block availability from COA and goods trains forecast	Pull data from 4 existing railway systems + predict freight train movement
2	Use AI/ML algorithms to prioritize and schedule based on criticality, urgency, and impact	ML model that scores which maintenance task is most urgent
3	Optimize block scheduling to maximize asset uptime, coordinating multi-department activities	OR-Tools solver that bundles Engg + S&T + Traction blocks together
4	Block plans over multiple time horizons — weekly and monthly	Dashboard showing 7-day tactical + 30-day strategic plans

Keywords Judges Will Look For:

- ✓ **"Goods trains forecast"** → explicitly asked for, most teams missed this
- ✓ **"Multi-department coordination"** → bundle blocks from 3 departments into one window
- ✓ **"Maximize asset availability"** → reduce how long the track is closed
- ✓ **"Weekly and monthly"** → must show multiple planning horizons

2. Understanding the Key Terms (Basics)

 TIP

If your team doesn't know these terms, the entire project will feel like a blackbox. Read this section first.

The 4 Systems (Data Sources)

SYSTEM	FULL NAME	DEPARTMENT	WHAT DATA IT HAS
TMS	Track Management System	Engineering (P-Way)	Track geometry defects, rail age, sleeper condition, GMT, USFD flaw reports
SMMS	Signalling Maintenance & Management System	Signal & Telecom (S&T)	Point machine health, track circuit failures, axle counter status, signal lamp faults
TDMS	Traction Distribution Management System	Electrical (TRD/OHE)	Overhead wire (catenary) wear, contact wire height, stagger values, SCADA transformer status
COA	Control Office Application	Train Operations	Real-time train running data, block corridor availability, section occupancy, passenger/goods schedules

Key Railway Concepts

TERM	SIMPLE EXPLANATION
Block	A reserved time window when a track section is closed for maintenance. No trains allowed.
Possession	The formal permission to occupy a track section for work. Same as "taking a block."
BDMS	Block Demand Management System — the existing CRIS-built software where departments request blocks. Our system sits ON TOP of BDMS.
Shadow Block	When dept A gets a block, dept B uses the same window for nearby work — "shadowing" on the same closure.
OHE	Overhead Equipment — the 25kV catenary wires powering electric trains. Must be switched off ("isolated") before TRD work.
GMT	Gross Million Tonnes — cumulative freight load a rail has carried. Higher GMT = rail needs replacement sooner.
TGI	Track Geometry Index — numeric score of how "out-of-shape" the track is. Lower TGI = urgent maintenance.
USFD	Ultrasonic Flaw Detection — detects internal cracks in rails invisible to the eye.
PTW	Permit to Work — formal safety document authorizing field staff to start maintenance.
TSR	Temporary Speed Restriction — speed limit on trains passing through a section under/after maintenance.
SLW	Single Line Working — when one track is blocked, trains run on the other track in both directions.

3. The Problem Explained Simply

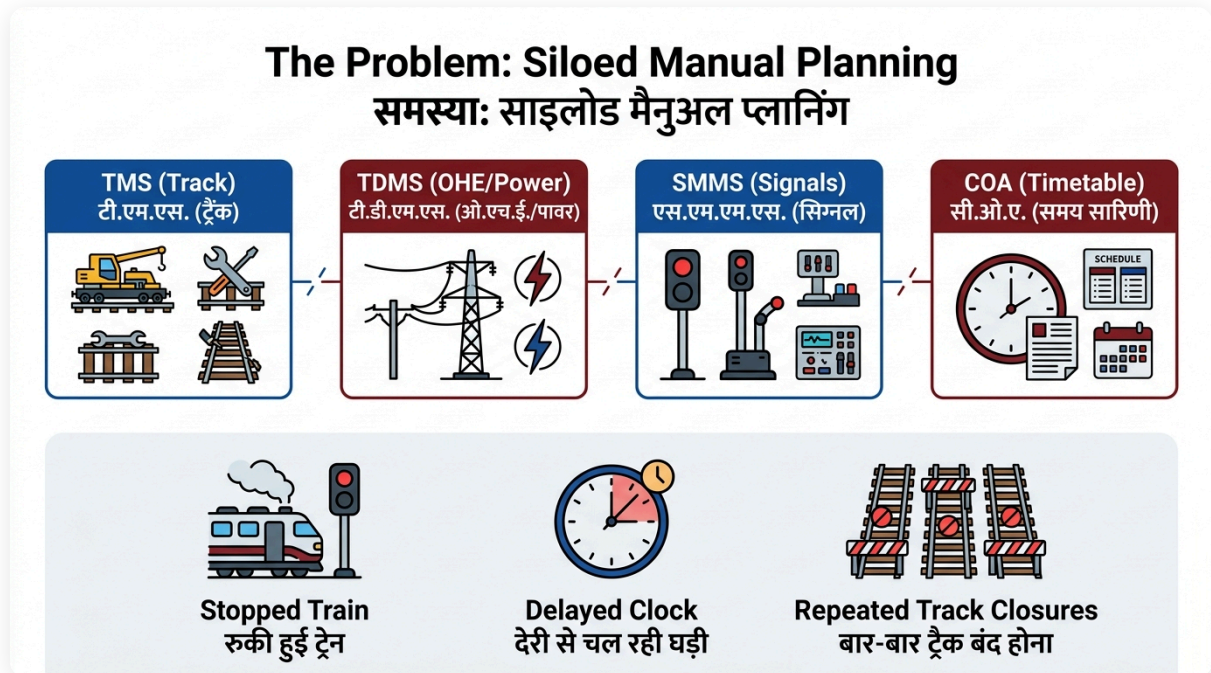


Figure 1: Currently, 4 systems operate in isolation — leading to repeated closures and delays

What's Happening Today?

Three departments (Engineering, S&T, Traction) all need to do maintenance on the **same track**. But they plan **independently**:

- 1 Monday:** Engineering blocks KM 127–128 for 3 hours to fix a sleeper.
- 2 Tuesday:** S&T blocks the **same section** for 2 hours to repair a point machine.
- 3 Wednesday:** TRD blocks the **same section** for 2.5 hours to fix OHE wire.

❌ RESULT

Track closed for
7.5 HOURS ACROSS 3 DAYS

when all three jobs could have been done in
ONE 4-HOUR WINDOW

What Does This Cause?

- ❌ **Repeated track closures** — same section blocked again and again
- 🕒 **Train delays** — Vande Bharat, Rajdhani, freight trains detained

- 💰 **Wasted resources** — crews mobilized 3 times instead of once
- ⚠️ **Safety risk** — more closures = more chances of human error

4. Competitor Analysis — 6 Teams Analyzed

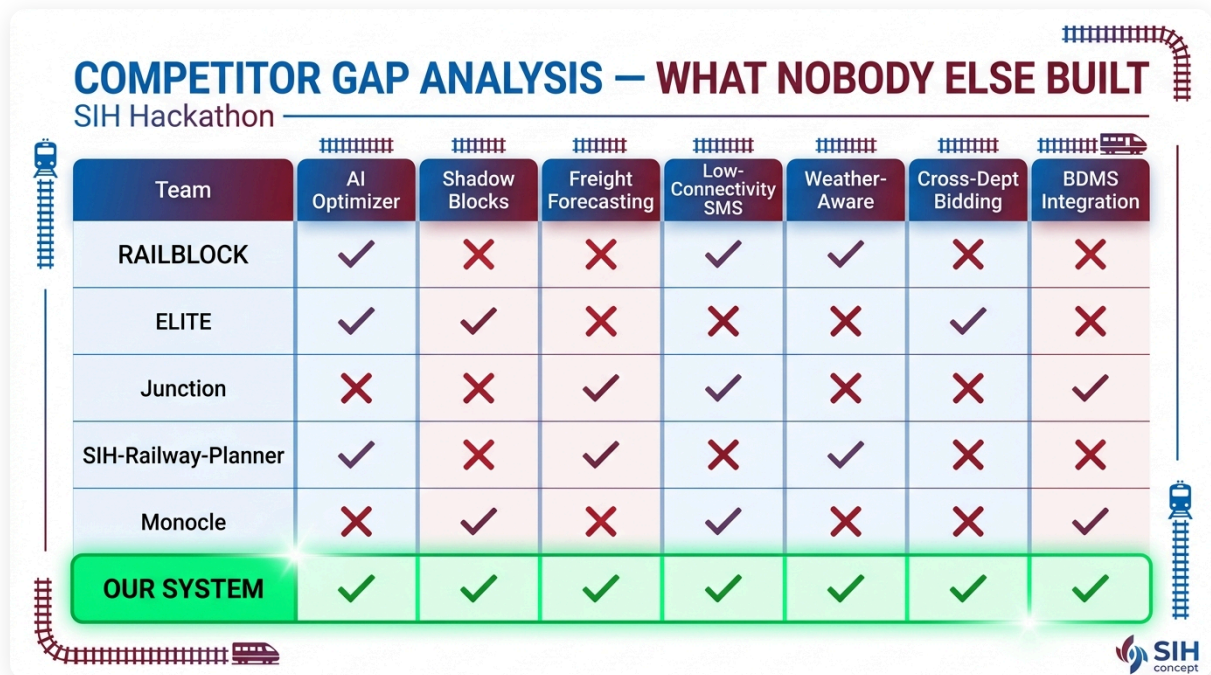


Figure 2: Feature gap analysis — what competitors have vs. what nobody built

Team 1: RAILBLOCK / RailPlan

Repo	sohambarik2007-pixel/RAILBLOCK-Indian-Railways-Design-System
Stack	Python backend, React frontend, data-gen module, Docker
Claims	OR-Tools CP-SAT, XGBoost + SHAP, GNN-based cascading delay prediction
Actually Built	3-phase lexicographic CP-SAT, scikit-learn GradientBoosting + Markov chains. GNN claim appears aspirational.
Strengths	Technically deepest optimizer, BDMS/COA field compatibility, what-if scrubber
Missing	✗ No freight forecasting · ✗ No weather · ✗ No low-connectivity · ✗ GNN may be oversold

Team 2: ELITE

Stack	PostgreSQL (Neon), JWT auth, OpenPyXL, LLM suggestions, Docker
Claims	"Self-learning AI + Railway Health Digital Twin + cross-department coordination"
Strengths	Human-in-the-loop approval, narrative packaging, validation before BDMS publishing
Missing	✗ "Digital Twin" is one-way visualization · ✗ No freight · ✗ No weather · ✗ No SMS. Impact numbers (11–17%) are self-estimated.

Team 3: JUNCTION ⚠️ (Strongest Competitor)

Repo	samyakmisal/junction Live: sihjunction.vercel.app
Stack	React 18 + TypeScript + Tailwind, FastAPI, Supabase (PostgreSQL + PostGIS), OR-Tools, XGBoost
Has	✓ MILP optimizer · ✓ Shadow Block clustering · ✓ Predictive maintenance (GMT/TGI/USFD) · ✓ XAI (SHAP) · ✓ Human-in-loop · ✓ What-If Sandbox · ✓ Multi-horizon (24H/7D/30D) · ✓ Audit trail · ✓ Dept workspaces · ✓ IR manual references (IRPWM, IRSEM, ACTM)
Missing	✗ No freight forecasting · ✗ No weather/monsoon · ✗ No low-connectivity SMS · ✗ No cross-dept bidding

Team 4: SIH-Railway-Planner

Repo	codes-from-neeraj/sih-railway-planner
Stack	Zero-dependency Python (stdlib only!), vanilla HTML/CSS/JS
Has	✓ IR Safety Compatibility Matrix (actual manual paragraphs: Para 808, ACTM Vol II Para 2063) · ✓ Decision audit (BUNDLE/SEQUENCE/DISRUPTION_SHIFT) · ✓ Disruption re-planner · ✓ Explicit scope disclosure · ✓ Track-Hours-Saved metric
Approach	Rule-based heuristic & greedy co-scheduling (explicitly discloses OR-Tools is for production scale-up)
Missing	✗ No actual ML model · ✗ No freight · ✗ No weather · ✗ No SMS · ✗ Simpler frontend

Team 5: MONOCLE (RailSync-AI)

Repo	AuraCode2007/Monocle
Stack	React 18 + Vite, FastAPI, OR-Tools CP-SAT, D3.js
Has	✔ CP-SAT optimizer · ✔ Joint co-located blocks · ✔ D3 String Chart (MARECHAL — IR standard) · ✔ Form T/348M PDF export (actual IR paperwork!) · ✔ What-If Simulator · ✔ Cryptographic Private Number · ✔ 26-week rolling horizon · ✔ "Ask RailSync" AI palette
Unique	Only team generating actual IR Form T/348M paperwork
Missing	✖ No freight forecasting · ✖ No weather · ✖ No SMS · ✖ No cross-dept bidding

5. What Every Competitor Already Has — Table Stakes

 **WARNING**

These features will NOT differentiate you. Every team has them. Build them as your foundation, but do NOT pitch them as innovation.

FEATURE	TEAMS THAT HAVE IT
Google OR-Tools CP-SAT optimizer	RAILBLOCK, Junction, Monocle
Multi-department shadow/bundled blocks	Junction, Monocle, SIH-Railway-Planner
ML priority scoring (XGBoost/GradientBoosting)	RAILBLOCK, Junction
Explainable AI (SHAP-style)	RAILBLOCK, Junction, Monocle
Human-in-the-loop approval	ELITE, Junction, SIH-Railway-Planner
Interactive Gantt timeline	All teams
What-If emergency simulator	Junction, Monocle, SIH-Railway-Planner
Multi-horizon planning (daily/weekly/monthly)	Junction, Monocle

Also table stakes globally: MTR Hong Kong's ETMS system already does AI-optimized possession planning in production (~51% efficiency gain). Network Rail (UK) funded the same in 2022. Optym sells it commercially to US railroads.

6. What NOBODY Has Built — Our Winning Edge

 **KEY INSIGHT**

These are the features that will make judges say "this team actually understands Indian Railways." Build these FIRST.

FEATURE	RAILBLOCK	ELITE	JUNCTION	PLANNER	MONOCLE	US ✓
Freight/Goods-train forecasting	✗	✗	✗	✗	✗	✓
Low-connectivity SMS/WhatsApp	✗	✗	✗	✗	✗	✓
Monsoon/Weather-aware rescheduling	✗	✗	✗	✗	✗	✓
Cross-department bidding mechanism	✗	✗	✗	✗	✗	✓

Every single one of these is unaddressed by all 5 competitor teams.

7. Final USP Priority Stack

USP #1 — TOP PRIORITY 🔥

Low-Connectivity Field Communication

What: Gangmen and JEs can submit block requests and confirm closures via **SMS / IVR (phone call) / WhatsApp**.

Why it wins: No competitor has it (0 out of 5 teams). Real Indian ground truth — many divisions have poor/zero 4G in remote track sections. A judge from Railway Board will immediately recognize this as genuine understanding of field conditions.

Implementation: Twilio SMS API → message format: `BLOCK REQ | SEC:NDL-GZB | KM:127-128 | DEPT:ENGG | DUR:3HR` → system parses into block request.

USP #2 — PS-MANDATED 🔥

Goods-Train / Freight Forecasting

What: Use real **data.gov.in freight loading data** to predict goods-train traffic per section, so the system knows which windows are actually free.

Why it wins: The PS literally says *"goods trains forecast from the Control Office."* Every competitor ignored this — they only look at passenger timetables (fixed) and ignore freight (dynamic, demand-driven).

Implementation: Download freight CSVs from data.gov.in → train time-series model (Prophet/ARIMA) → feed predicted windows as optimizer constraints.

USP #3

Monsoon / Weather-Aware Planning

What: Pull **IMD weather alerts** and auto-reschedule blocks around flood-prone sections, cyclone paths, and heatwave periods.

Why it wins: India-specific operational reality nobody touches. Easy to demo: show "before alert" vs "after alert" auto-adjusted schedule.

Implementation: IMD RSS/API feed → parse warnings by region → flag affected KM sections → optimizer marks windows unavailable → auto-reschedules.

USP #4 — NICE TO HAVE

Cross-Department Bidding Mechanism

What: Departments "bid" for block time on the same section, AI bundles compatible bids into shared possessions.

Priority: Build LAST. Cut FIRST if running short on time.

LAYER	WHAT	PITCH AS
Foundation	BDMS-compatible integration layer (real field names: PTW, Private Numbers, COA codes)	Present as base infrastructure, NOT as innovation
Demo/UX	Interactive map visualization	Polish for demo day — helps presentation but not a USP

8. Complete System Workflow — 8 Steps

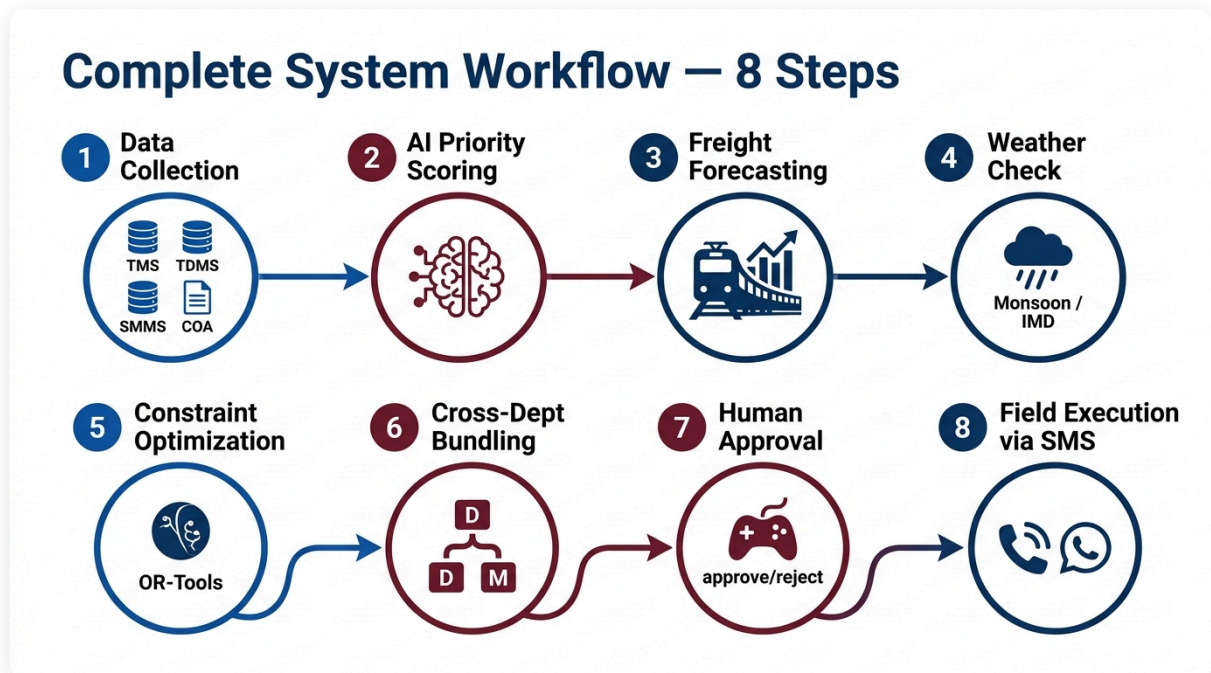


Figure 3: End-to-end workflow from data collection to field execution

- 1 Data Collection & Ingestion:** Pull maintenance tasks, defects, and overdue items from TMS, SMMS, TDMS. Pull train timetable and block corridor availability from COA. Normalize into a unified task database. Each task: location (KM range), department, work type, duration, urgency.
- 2 AI Priority Scoring:** ML model (XGBoost) scores each task on Criticality (USFD crack=critical, routine tamping=low), Urgency (past SOP deadline=high), Impact (mainline section=high). Output: Priority score 0–100.
- 3 Freight Forecasting 🔥 (OUR USP):** Time-series model on data.gov.in freight data. Predicts goods-train volume per section per day. Marks "freight-heavy windows" as constraints for the optimizer.
- 4 Weather Check 🔥 (OUR USP):** Pull IMD weather alerts. Flag affected KM sections. Mark flooded/unsafe sections as "no-work zones." Auto-reschedule affected blocks to safe windows.
- 5 Constraint Optimization (OR-Tools CP-SAT):** Multi-objective: minimize train delays, maximize asset availability, maximize cross-dept bundling. Constraints: safety rules, OHE isolation, headway, crew availability. Output: Optimal block schedule.
- 6 Cross-Department Bundling:** Identify tasks from different departments targeting the same KM section. Check safety compatibility. Bundle into "shadow blocks." Calculate Track-Hours-Saved metric.

7

Human-in-the-Loop Approval: Chief Controller reviews AI plan. Options: Approve  / Modify Time Window  / Reject . XAI panel shows WHY each slot was chosen. Modified blocks re-solve.

8

Field Execution via SMS/WhatsApp  (OUR USP): Approved blocks sent to field staff via SMS. Gangman confirms start/completion via reply. System logs actual vs. planned for feedback loop.

9. Technical Architecture & Stack

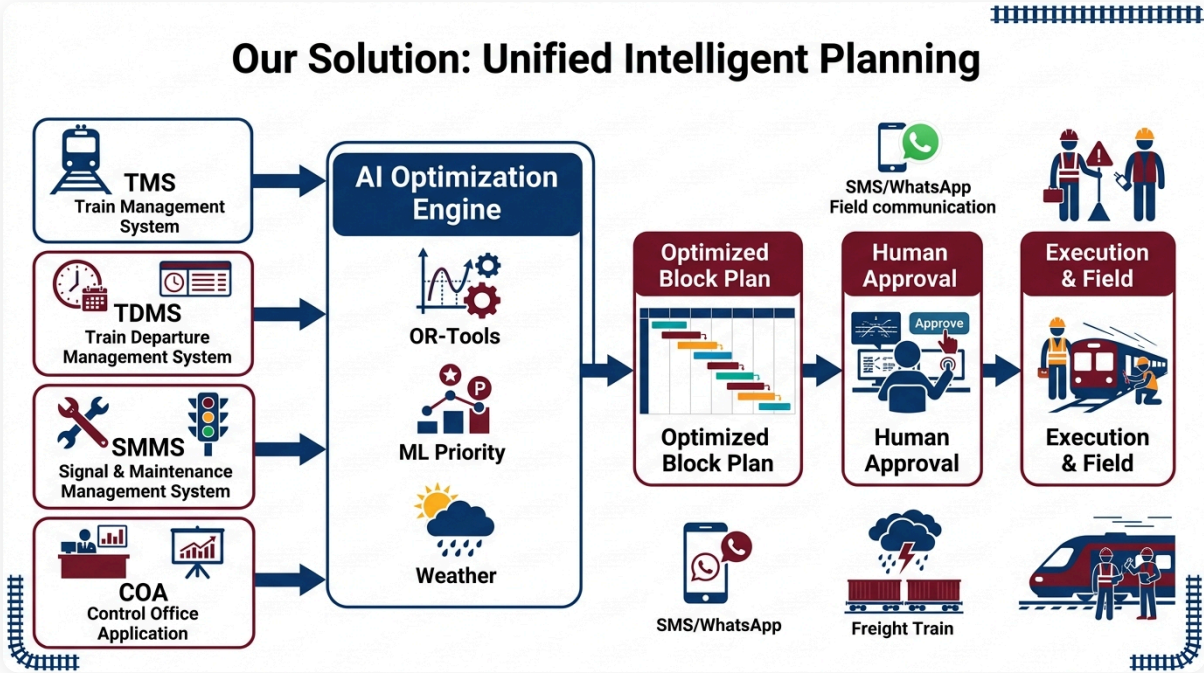


Figure 4: Unified architecture — from siloed data to coordinated field execution

Recommended Tech Stack

LAYER	TECHNOLOGY	WHY
Frontend	React 18 + Vite + Tailwind CSS	Fast, modern, all competitors use it
Backend	Python FastAPI	Async, auto-docs (Swagger), lightweight
Database	PostgreSQL (Supabase / Neon)	Free tier, PostGIS for spatial KM data
Optimizer	Google OR-Tools CP-SAT	Free, proven, industry standard
ML	scikit-learn (XGBoost)	Simple, fast, enough for priority scoring
Freight Forecast	Prophet / ARIMA	Time-series on data.gov.in freight data
Weather	IMD API / RSS feed	Free, real Indian weather data
SMS/WhatsApp	Twilio API	Free trial, easy integration
Charts	Recharts / D3.js	Gantt timelines, MARECHAL string charts
Deploy	Vercel + Railway.app	Free tiers, instant deploy

10. Winning Pitch Script

Opening Hook (15 seconds)

*"Right now, Engineering, Signal & Telecom, and Traction departments plan maintenance blocks independently through BDMS — a real, already-deployed CRIS system. But the planning on top of it is still manual, disconnected, and blind to real ground conditions. That's not a hypothetical gap — **it's the exact bottleneck the Ministry named in this problem statement.**"*

Positioning (10 seconds)

*"AI-optimized possession planning already exists — MTR Hong Kong and Network Rail have it. So our differentiator isn't 'we used AI.' **It's that we built the only solution designed for how Indian Railways actually operates on the ground.**"*

USP Walkthrough (60 seconds)

USP 1: *"Our system talks to the field the way the field actually works — gangmen and JEs in low-connectivity sections can submit block requests and closures over SMS/WhatsApp, not just a web dashboard."*

USP 2: *"We don't assume the corridor is free — we forecast it. Using real Indian Railways freight-loading data from data.gov.in, we predict goods-train movement to know which windows are actually available. The PS explicitly asks for this. No other team built it."*

USP 3: *"We plan around India's own risks — monsoon floods, cyclones, heatwave speed restrictions. IMD weather alerts automatically reschedule unsafe blocks."*

USP 4: *"Departments bid for block time cooperatively — the AI bundles compatible bids into shared possessions."*

Closing (10 seconds)

*"Every other team building this is competing on optimization math. **We're competing on operational reality — and that's what a railway safety evaluator will actually trust.**"*

⚡ KEY RULES FOR THE TEAM

1. **DO NOT spend time polishing the optimizer** — that's table stakes. Every team has OR-Tools.
2. **DO spend all remaining time on USPs #1, #2, #3** — that's where we win.
3. **DO NOT claim numbers you can't prove** — other teams cite "51.3% reduction" from synthetic data. Run YOUR model, report YOUR numbers.
4. **DO reference real IR manuals** — IRPWM, ACTM Vol II, IRSEM. This signals domain depth.
5. **USP #4 (Cross-dept bidding)** — build LAST, cut FIRST if short on time.

Document prepared: September 7, 2026 | Based on analysis of 6 competitor teams + official PS
SIH26027 + global railway AI systems
For internal team reference only.